JAMES MATTHEW HARRISON

Principal Engineer

Email: james.harrison@gmail.com

Phone: +44 7892 445623

LinkedIn: linkedin.com/in/jamesharrison-engineer

PROFESSIONAL SUMMARY

I'm a Principal Engineer with 14 years of experience designing and delivering large-scale distributed systems across financial services and SaaS environments. I've led technical strategy for teams ranging from 5 to 25 engineers, mentoring staff-level engineers and establishing architectural standards that have reduced deployment times by 40 percent. My focus has been on building resilient infrastructure and guiding organisations through complex technology transitions while maintaining system reliability. I enjoy solving problems at the intersection of scalability, security, and team productivity.

WORK EXPERIENCE

Principal Engineer

Catalyst Systems

https://www.catalyst-systems.io

Edinburgh, United Kingdom

March 2021 - Present

Led architectural decisions for a microservices platform serving 150 million daily transactions across UK and EU markets, establishing standards that reduced system outages by 65 percent year-on-year.

Mentored eight engineers across infrastructure and backend teams, with three promoted to senior roles during my tenure by developing their architectural thinking and decision-making capabilities.

Implemented observability framework using Prometheus and ELK stack that reduced incident response time from 45 minutes to 12 minutes, improving customer-facing SLA compliance to 99.98 percent.

Designed and executed migration from monolithic .NET application to Kubernetes-based architecture, completing the transition across 18 services with zero customer impact and training 22 engineers on container orchestration practices.

Senior Software Engineer

Meridian Financial Solutions

https://www.meridian-financial.com

London, United Kingdom

June 2017 - February 2021

Architected distributed payment processing system handling GBP 4.2 billion in annual transaction volume using event-driven design patterns, achieving 99.99 percent uptime over three years.

Built and scaled a platform team of 12 engineers responsible for shared infrastructure, establishing deployment pipelines and CI/CD practices that reduced release cycles from three weeks to two days.

Led technical due diligence and integration efforts following acquisition of competing fintech company, consolidating three separate systems into unified architecture within eight months.

Introduced automated performance testing frameworks that identified critical bottlenecks in payment validation logic, resulting in 35 percent improvement in transaction processing latency.

Software Engineer

Vertex Technology Group

https://www.vertextech.net

Edinburgh, United Kingdom

July 2010 - May 2017

Developed backend services in Java and C# for SaaS CRM platform used by 5,000 enterprise customers, contributing to revenue growth from GBP 2 million to GBP 18 million.

Established database optimisation practices that reduced query response times from 8 seconds to 0.8 seconds across core reporting features, significantly improving user experience for large customer deployments.

Led adoption of REST API standards and implemented comprehensive API versioning strategy that reduced integration issues for partner companies by 72 percent.

Contributed to security hardening initiatives following external audit, implementing OAuth 2.0 authentication, data encryption at rest, and role-based access controls across the platform.

EDUCATION

BSc Computer Science

University of Edinburgh

Edinburgh, United Kingdom

Graduated: June 2010

SKILLS

Languages: Java, C#, SQL, Python, Go, Bash

Cloud and Infrastructure: AWS, Azure, Docker, Kubernetes, Terraform, CloudFormation

Databases: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch

Message Queues: RabbitMQ, Apache Kafka, AWS SQS

Monitoring and Observability: Prometheus, Grafana, ELK Stack, DataDog, New Relic

Architecture: Microservices, Event-Driven Architecture, CQRS, Domain-Driven Design, Distributed Systems

DevOps and CI/CD: Jenkins, GitLab CI, GitHub Actions, ArgoCD, Helm

Soft Skills: Technical Strategy, Team Leadership, Mentoring, Architecture Review, Cross-functional Collaboration